

Animal Image Classification with Convolutional Neural Networks

Artificial Neural Networks 2025/2026

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Problem Statement

Animal image classification is a fundamental computer vision task with applications in wildlife monitoring, biodiversity research, veterinary diagnostics, and educational tools. In this exercise, you will design, train, and evaluate a Convolutional Neural Network (CNN) capable of automatically classifying images of animals into their correct categories based on visual features.

You are provided with an image dataset containing photographs of different animal species, organized by class labels.

The dataset includes:

- Multiple animal categories (each category represented as a separate class)
- RGB images of varying sizes and backgrounds
- Labeled data suitable for supervised learning

The dataset is structured as:

- **Training images:** (from data/train/): used to learn visual patterns such as shapes, textures, and colors. In the subfolder bears/ and in elephants/ you have 200 images each.
- **Test images:** (from data/test/): used **once** at the end to evaluate generalization on unseen data. In the subfolder bears/ and in elephants/ you have 20 images each.

Each image:

- Represents a single animal
- Belongs to exactly one class
- May vary in pose, lighting conditions, background, and scale

The goal is to build an image classification model using convolutional neural networks that can accurately predict the animal class for unseen images. The exercise will involve data preprocessing, CNN architecture design, model training, and performance evaluation using appropriate classification metrics.

Operational Evaluation

Q1. Baseline Linear Classifier (Flatten $\rightarrow$ Linear $\rightarrow$ Softmax). Resize images to 128×128 , normalize RGB, flatten, and train a single Linear layer $\rightarrow$ num_classes with CrossEntropyLoss. Report top-1 accuracy on the validation split.

Q2. Create a CNN with One Convolutional Block. Resize to 128×128 , normalize; model: Conv2d($3 \rightarrow 32$, 3×3 , padding=1) $\rightarrow$ MaxPool2d(2×2) $\rightarrow$ Flatten $\rightarrow$ Linear(num_classes) $\rightarrow$ Softmax. Keep the same preprocessing as Q1.

Q3. Add Activation Functions to the Conv Block. Insert ReLU after the convolution: Conv2d($3 \rightarrow 32$, 3×3 , padding=1) $\rightarrow$ ReLU $\rightarrow$ MaxPool2d(2×2) $\rightarrow$ Flatten $\rightarrow$ Linear (to num_classes) $\rightarrow$ Softmax.

Q4. Forward Pass Classification. Use CrossEntropyLoss and SGD or Adam. Run a single forward/backward step on one mini-batch, confirm loss decreases on a few successive steps and that gradients/parameters update as expected.

Q5. Train the CNN and Track Learning Curves. Train the Q3 model for 5 epochs; log train/val loss and accuracy per epoch; plot loss & accuracy vs. epochs; report the best validation accuracy.

Q6. Evaluate with Classification Metrics. On the val/test split, compute Top-1 accuracy, macro Precision/Recall/F1, and a confusion matrix. Report per-class F1 and highlight the three weakest classes.

Q7. Tiny Hyperparameter Grid. Grid: filters $\in \{32, 64\}$, lr $\in \{1e-3, 3e-4\}$, batch $\in \{64, 128\}$, weight_decay $\in \{0, 1e-4\}$. Train 3 epochs/config; select the model with the best macro-F1 on validation.

Q8. Visual Diagnostics of Predictions. (a) Plot a row-normalized confusion matrix heatmap, (b) Show a 4×4 grid of validation images (8 correct, 8 incorrect) with true vs. predicted labels and predicted probabilities. Add brief notes on failure patterns (e.g., background bias, pose, scale).

Q9. Inference Deliverable (Top-k Predictions) With the best model, run on a folder of unseen images (same preprocessing). Save a CSV with filename, top-1 prediction, and top-5 predictions with probabilities. Report throughput (img/s) on CPU/GPU.

Q10. Explainability & Error Analysis. Show a small grid (e.g., 3×3) with 5 correct and 4 incorrect validation images. For each, display true vs. predicted and prediction confidence. Write 3 brief observations about failure patterns (e.g., background, pose, scale).